

CSE2315 — Assignment 5

Vlad Paun
6152937

March 16, 2026

Exercise 1

Consider the alphabet $\Sigma = \{a, b, c\}$ and the language

$$L = \{a^k b^k c^k \mid k \geq 1\}.$$

Give a high-level description for a three-tape enumerator that enumerates L . You may want to use the second and third tapes as counters. You do not have to give implementation details of the counters.

Answer: Use Tape 1 as the printer and Tapes 2 and 3 as counters. Tape 2 stores the current value of k , initially $k = 1$.

In each iteration, the machine copies k from Tape 2 to Tape 3 and uses Tape 3 to print k copies of a , b , and c on the printer tape, producing the string $a^k b^k c^k$. It then increments k on Tape 2.

Repeating this process prints $abc, aabbcc, aaabbbccc, \dots$, thus enumerating all strings in $L = \{a^k b^k c^k \mid k \geq 1\}$.

Exercise 2

Old exam question endterm 23–24:

- (a) **Give a Turing Machine M that is not a decider such that $L(M)$ is decidable.**
- (b) **Explain why M is not a decider.**
- (c) **Explain why $L(M)$ is decidable.**

Exercise 3

Old exam question resit 20–21: Let

$$MIA_{TM} = \{\langle M_1, M_2 \rangle \mid L(M_1) \cap L(M_2) \neq \emptyset\}.$$

Show that this is a recognizable language.

Exercise 4

In the game of chess, consider an intermediate board configuration, and suppose it is white's move. We know that white is either guaranteed to win or not. Let

$L = \{w \mid w \text{ represents a board configuration, and white is guaranteed to win if it is white's move and white plays optimally}\}$

Is L recognizable, decidable, or neither? Explain.

Exercise 5

Revision, old exam question, midterm 21–22: Given a DTM M with $\Sigma = \{a, b\}$ such that:

- M has no transition to its q_{reject} , i.e. the reject state is an unreachable state,
- $L(M) = \{w \in \Sigma^* \mid w \text{ starts with an } a\}$.

Is M a decider, not a decider, or can we not conclude that? Explain your answer.

Exercise 6

Old exam question resit 20–21:

In a log of a scheduler, it is noted down when a request for a task is given, and when an action is done to do that task. Requests are noted down with r , actions with a . A log is a final log if all tasks have been done. Thus, a final log can be described by the language of words where every r has a corresponding a that appears later in the string, formally given as

$L = \{w \mid w \text{ has an equal amount of } r\text{'s and } a\text{'s and the } i\text{th } a \text{ appears later in the string than the } i\text{th } r, \text{ for each } i\}$

Give an informal description of a Turing machine with the alphabet $\{r, a\}$ that decides this language. Use a maximum of 15 lines.

As an example:

$$\text{rarraa} \in L \quad \text{and} \quad \text{raaarr} \notin L$$

Exercise 7

Revision, old exam question midterm 19–20: Consider the following rules R of a CFG

$$G = (V, \Sigma, R, S),$$

with

$$V = \{S, A, B\} :$$

$$S \rightarrow ASA \mid Ba$$

$$A \rightarrow bB \mid S$$

$$B \rightarrow c \mid b \mid \varepsilon$$

- (a) Give a valid set Σ such that $|\Sigma| = 5$.
- (b) Convert G into Chomsky Normal Form (CNF). Use the procedure from Sipser, showing intermediate steps.